

THE UNIVERSITY OF EDINBURGH

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Assignment A1: SmeltCo Ltd Monthly Production Plan

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Introduction

SmeltCo Ltd is a company that produces three types of metal alloys, including brass, bronze, and duralumin, through blending various raw materials such as copper, zinc, tin, and aluminium. The company is looking for an optimal monthly production plan that will maximize profit, taking into consideration the limitations in raw material availability and production capacity, as well as ensuring good quality alloys are produced by meeting the composition requirements of each alloy. Furthermore, 10% of all input materials are lost during the blending process.

The primary objective of this study is to determine an optimal production quantity of brass, bronze, and duralumin that will maximize SmeltCo Ltd.'s monthly profit. The study will employ linear optimization modelling to indicate the most profitable production mix and identify the raw materials that act as the binding constraints on production and profitability. The decision variables in this optimization problem will be the production quantities of each alloy and the corresponding input quantities of raw materials (all in kilograms).

Relevance of the Problem

As SmeltCo operates under limited resources due to the scarce availability of raw materials, the company needs a linear optimization model that will allow it to make data-driven decisions on the right mix of alloys that should be produced and the most critical quantity of raw materials that should be purchased in the respective month.

Beyond SmeltCo, the study demonstrates the real-world practical application of linear optimization in industrial operations. Through the application of mathematical modelling techniques, the companies can improve their resource utilization and maximize profits.

Assumptions

The modelling will take into consideration the following assumption.

- i. Demand for all products is sufficient to sell everything produced.
- ii. Raw material prices and selling prices are constant
- iii. There are no storage or transportation constraints.

Modelling Approach

To solve SmeltCo's decision-making problem of determining the optimal monthly production plan, the linear optimization model is formulated. The model represents the company's key production and resource utilization decisions using decision variables, an objective function, and constraints that include the availability of raw materials, product quality, and monthly production capacity, as outlined below. The main objective is to maximize profit while satisfying all the constraints.

Set and Parameters

Symbol	Description
P	Set of alloy products produced, $P = \{\text{Brass, Bronze, Duralumin}\}$
M	Set of raw materials used in production, $M = \{\text{Aluminium, Copper, Tin, Zinc}\}$
S_p	Selling price (£/kg) of alloy product, $\forall p \in P$
C_m	Purchase cost (£/kg) of raw material, $\forall m \in M$
Q_m	Maximum purchase quantity (kg) available for raw material, $\forall m \in M$
PR_p	Maximum monthly production capacity (kg) for alloy, $\forall p \in P$
MU	Material usage proportion (1- Material loss (10%) rate), $ML \in R$
$MMin_{mp}$	Minimum required proportion of raw material (m) in product (p), $\forall m \in M, p \in P$
$MMax_{mp}$	Maximum required proportion of raw material (m) in product (p), $\forall m \in M, p \in P$

Decision Variables

Let our Decision variables be:

Symbol	Description
X_p	Kilograms of alloy products produced monthly, $\forall p \in P$
Y_{mp}	Kilograms of materials used in producing a given type of alloy product, $\forall m \in M, p \in P$

Objective Function

The main goal is to maximize the total profit, which is given by the objective function below.

$$\text{Maximize } (Z) = \sum_{\{p \in P\}} S_p \cdot X_p - \sum_{\{m \in M\}} C_m \cdot \left(\sum_{\{p \in P\}} Y_{mp} \right)$$

Constraints

- i. Material loss constraint

We are told that 10% of the materials were lost during the blending process; therefore, the alloy products were actually made from 90% of all the materials used to produce different alloy types (i.e., $MU = 0.9$). We can write this constraint as;

$$X_p = MU \cdot \sum_{\{m \in M\}} Y_{mp} \quad \forall p \in P$$

- ii. Production Capacity Constraint

Due to capacity restrictions, SmeltCo Ltd can produce no more than 70000 Kg of brass, no more than 100000 Kg of bronze, and no more than 90000 Kg of duralumin. Therefore, the production amount for each alloy product is limited by the given capacity restrictions.

$$X_p \leq PR_p \quad \forall p \in P$$

iii. Raw Material Availability Constraint

The total amount of the quantity of raw material available to make an alloy product for a given month must not exceed the maximum monthly purchase quantity.

$$\sum_{\{p \in P\}} Y_{mp} \leq Q_m \quad \forall m \in M$$

iv. Proportion of Required Alloy Composition Constraint

To ensure that SmeltCo maintains the quality of the alloys produced, we have been provided with the minimum and maximum proportions of material mix for each product. The values for $MMin_{mp}$ and $MMax_{mp}$ will be zero (0) if the material is not involved in a product. The constraints for the minimum and maximum requirements will be as shown below:

$$Y_{mp} \geq MMin_{mp} \cdot X_p / MU \quad \forall m \in M, p \in P$$

$$Y_{mp} \leq MMax_{mp} \cdot X_p / MU \quad \forall m \in M, p \in P$$

v. Blending Requirement

We will also add the blending requirement to show the required mix for each alloy in a matrix form as an array of materials and products.

$$\sum_{\{m \in M\}} Y_{mp} = X_p / MU \quad \forall p \in P$$

vi. Non-negative (Domain) constraint

We also need to add non-negative constraints, that is to say, the kilograms of alloys produced and raw material purchased must not be less than zero.

$$X_p \geq 0, Y_{mp} \geq 0 \quad \forall m \in M, \forall p \in P$$

The General Linear Programming Model will be:

Maximize:

$$\sum_{\{p \in P\}} S_p \cdot X_p - \sum_{\{m \in M\}} C_m \cdot \left(\sum_{\{p \in P\}} Y_{mp} \right)$$

$$\text{s.t. } X_p = MU \cdot \sum_{\{m \in M\}} Y_{mp} \quad \forall p \in P$$

$$X_p \leq PR_p \quad \forall p \in P$$

$$\sum_{\{p \in P\}} Y_{mp} \leq Q_m \quad \forall m \in M$$

$$Y_{mp} \geq MMin_{mp} \cdot X_p / MU \quad \forall m \in M, p \in P$$

$$Y_{mp} \leq MMax_{mp} \cdot X_p / MU \quad \forall m \in M, p \in P$$

$$\sum_{\{m \in M\}} Y_{mp} = X_p / MU \quad \forall p \in P$$

$$X_p \geq 0, Y_{mp} \geq 0 \quad \forall m \in M, p \in P$$

Results

Summary of the Problem

SmeltCo Ltd produces three metal alloys, brass, bronze, and duralumin, by blending different raw materials under specific quality and operational constraints. Each alloy has a distinct composition requirement to ensure product quality: brass must contain 20 to 50% copper, bronze 60 to 80% copper, and duralumin at least 45% aluminium and 10 to 50% copper. The company faces monthly production capacity limits of 70,000 kg for brass, 100,000 kg of bronze, and 90,000 kg for duralumin, and incurs a 10% material loss during the blending process.

SmeltCo purchases four raw materials, aluminium (max 50,000 kg at £8/kg), copper (max 80,000 kg at £12/kg), tin (max 75,000 kg at £2/kg), and zinc (max 100,000 kg at £5/kg) and sells the finished alloys at £8/kg for brass, £16/kg for bronze, and £14/kg for duralumin. The optimization model was formulated to determine the combination of alloy production levels that maximizes total monthly profit, while satisfying all quality, capacity, and material availability constraints.

The results presented below show the optimal production plan that meets these operational restrictions and the profit-maximizing use of each raw material, together with an analysis of which constraints are most critical to overall profitability.

Overview of the Results

The linear programming model was solved using Xpress. The optimal solution obtained indicates that SmeltCo should produce 35,000 Kg of brass, 100,000 Kg of bronze, and 50,000 kg of duralumin per month to maximize the overall monthly profit. The maximum profit obtained under this production mix is **£975,555.56**.

The summary of the results obtained for both the quantity of alloys that need to be produced per month and the kilograms of materials used for each alloy is shown in Tables 1 and 2 below.

Table 1: Optimal production capacity

Product	Optimal production (kg)	Unit selling price (£/kg)	Revenue (£)
Brass	35,000	8	280,000.00
Bronze	100,000	16	1,600,000.00
Duralumin	50,000	14	700,000.00

Table 2: Product material requirements

Material Used	Brass	Bronze	Duralumin	Total Amount Used (kg)
Aluminum			50,000.00	50,000.00
Copper	7,777.78	66,666.67	5,555.56	80,000.00
Tin		44,444.44		44,444.44
zinc	31,111.11			31,111.11

Interpretation of Results

From the results, we can see that bronze reaches its production capacity limit of 100,000 kg per month. If the company could have the capacity to produce additional units of bronze, it would further increase the profit.

Brass is produced at an intermediate level of 35,000 kg, which balances the use of copper and zinc, since copper is first allocated to bronze and duralumin, as they generate higher revenue compared to brass. The production of duralumin is limited by the availability of aluminium, showing that aluminium is a key material for the organization.

Another observation from the results is that the production mix obtained shows that the model ensures that the scarce materials are allocated to the most profitable products.

Sensitivity Analysis

Sensitivity analysis was conducted to determine how much profit would increase if an additional kilogram of each raw material were available. The model was run by increasing each material constraint by one kilogram and observing the changes in the optimal value (shadow price). The shadow prices (dual values) indicate the marginal worth of relaxing each material constraint. The summary for the sensitivity analysis is presented in Table 3 below.

Table 3: Summary of sensitivity analysis (Material shadow prices)

Material	Shadow Price (£/kg)	Binding Status	Strategic Implications
Aluminium	4.22	Binding	It is worth paying up to £4.22 more per kg
Copper	4	Binding	It is worth paying up to £4.00 more per kg
Tin	0	Not binding	Extra supply adds no value
Zinc	0	Not binding	Extra supply adds no value

The analysis above shows that:

- Aluminium has a shadow price of £4.22/kg, meaning that increasing the aluminium supply by 1 kg would raise total profit by £4.22, up to the point where the other constraint takes over and becomes binding.
- Copper has a shadow price of £4.00/kg, implying a £4.00 increase in profit per extra kilogram of copper supply.
- Tin and Zinc have shadow prices of £0, indicating that their availability is not scarce and therefore is not currently limiting profit.

Managerial Implications and Strategic Recommendations

The optimization results provide clear guidance for SmeltCo Ltd.'s production and procurement strategy. If the company has an opportunity to negotiate with the suppliers, it should consider the following.

- Prioritize negotiating for the extra supply of aluminium as it is one of the binding resources and directly determines the achievable profit, provided the additional cost per kg is not more than £4.22.
- Consider negotiating for increased copper supply, provided the additional cost per kg is below £4.
- Use the extra aluminium and copper to produce more duralumin, as it offers high profit at a high selling price, and the company is capable of producing 40,000 kg more of duralumin per month if the resources are available.
- Review bronze production capacity, as increasing capacity could further improve profitability, as bronze production currently reaches its maximum allowable limit of 100,000 kg.
- Avoid over-purchasing tin and zinc, as these materials are not fully utilized under optimal conditions.

Conclusion

The linear optimization model successfully determined SmeltCo Ltd.'s optimal monthly production plan, yielding a maximum profit of £975,555.56. This plan utilizes the full production capacity for bronze and is constrained by the limited availability of certain raw materials to reach the full production capacity for brass and duralumin.

Our sensitivity analysis confirms that aluminium and copper are the most critical bottleneck resources, as evidenced by their positive shadow prices of £4.22/kg and £4/kg respectively. We strongly recommend that SmeltCo Ltd prioritize negotiating an increased supply of these materials. The company can justify paying an extra price of £4.22/Kg for aluminium and £4/kg for copper (the current shadow prices) for additional supplies, as any price below this threshold will result in a net increase in overall monthly profit.